

Stack Overflow Developer Survey Analysis

Project Overview

This project performs comprehensive statistical analysis on the Stack Overflow Developer Survey dataset using Python and Pandas. The analysis explores developer demographics, work preferences, programming language popularity, compensation patterns, and educational backgrounds across various segments of the developer community.

Objectives

- Practice working with large datasets using Pandas
- Apply descriptive statistics and data aggregation methods
- Analyze measures of central tendency and percentage distributions
- Perform grouping and filtering operations with multiple criteria
- Handle missing values and work with sets
- Visualize analysis results in clear tabular formats

Tech Stack

- Programming Language: Python 3.x
- Primary Library: Pandas
- Environment: Jupyter Notebook / Google Colab
- Data Source: Stack Overflow Developer Survey (latest available)

Dataset Structure

The analysis uses two main files:

1. `survey_results_public.csv`: Main survey responses
2. `survey_results_schema.csv`: Data dictionary containing column descriptions
 - **Field qname**: Question names/identifiers

Project Tasks

Task 1: Total Respondent Count

Goal: Determine the total number of survey participants

Skills Applied:

- **DataFrame loading**
- **Record counting methods**
- **Basic data exploration**

Output: Single numeric value representing total respondents

Task 2: Response Completeness Analysis

Goal: Identify respondents who answered all survey questions

Skills Applied:

- **Loading schema file to extract question names**
- **Set intersection to filter relevant columns**
- **Missing value detection**
- **Complete case analysis**

Output: Count of respondents with no missing values across all survey questions

Task 3: Work Experience Statistical Analysis

Goal: Calculate measures of central tendency for the WorkExp field

Metrics:

- **Mean (average)**
- **Median (50th percentile)**
- **Mode (most frequent value)**

Skills Applied:

- **Descriptive statistics**
- **Handling missing values**
- **Statistical measure calculation**

Output: Table/series showing all three central tendency measures

Task 4: Remote Work Analysis

Goal: Count respondents working remotely

Skills Applied:

- **Column identification**
- **Data filtering**
- **Value counting**

Output: Numeric count of remote workers

Task 5: Python Popularity Assessment

Goal: Calculate the percentage of respondents who program in Python

Skills Applied:

- Text data manipulation
- Handling multiple-value fields (comma-separated languages)
- Percentage calculation
- String matching/contains operations

Output: Percentage value (e.g., 45.3%) of Python programmers

Task 6: Learning Path Analysis

Goal: Determine how many respondents learned programming through online courses

Skills Applied:

- Categorical variable filtering
- Multiple-response field handling
- Boolean indexing

Output: Count of respondents who used online courses for learning

Task 7: Geographic Compensation Analysis (Python Developers)

Goal: Analyze compensation patterns for Python developers by country

Metrics per Country:

- Mean annual compensation (ConvertedCompYearly)
- Median annual compensation

Skills Applied:

- **Multi-condition filtering (Python programmers only)**
- **GroupBy operations**
- **Aggregation functions**
- **Missing value handling in numeric fields**

Output: DataFrame with columns: Country, Mean Compensation, Median Compensation

Task 8: Education Level Analysis (Top Earners)

Goal: Identify education levels of the top 5 highest-paid respondents

Skills Applied:

- **Data sorting (descending by compensation)**
- **Top-N record selection**
- **Column extraction**

Output: List/table of 5 education levels from highest earners

***Task 9: Python Popularity by Age Group (Optional)**

Goal: Calculate Python adoption percentage across different age categories

Skills Applied:

- **Age-based grouping**
- **Within-group percentage calculation**
- **Cross-tabulation analysis**

Output: Table with columns: Age Category, Python Developer Percentage

***Task 10: Industry Analysis (High-Earning Remote Workers) (Optional)**

Goal: Identify most common industries among high-earning remote workers

Criteria:

- **Compensation $\geq$ 75th percentile**
- **Remote work status = True**

Skills Applied:

- **Percentile calculation**
- **Multiple condition filtering (Boolean AND)**
- **Frequency counting**
- **Descending sort by popularity**

Output: Ranked list/table of industries with their frequencies

Key Skills Demonstrated

- **Data Loading & Exploration: Working with large CSV datasets**
- **Data Cleaning: Handling missing values, filtering invalid data**
- **Statistical Analysis: Mean, median, mode, percentiles**
- **Data Manipulation: Filtering, grouping, aggregating, sorting**
- **Set Operations: Intersection for column filtering**
- **String Operations: Text matching, substring detection**

- **Aggregation: GroupBy operations with multiple metrics**
- **Complex Filtering: Multi-criteria Boolean indexing**
- **Data Presentation: Clear tabular output formatting**